



CASE STUDY

Transforming Data Integrity into Business Value

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NOTE

This is an independent case study using public data.

All results are simulated to demonstrate my analytical approach.

Problem:

IBM built an AI tool to predict employee would quit with **87% accuracy**.

Is this AI tool safe, fair, and ready to use?

Solution:

A 4-step audit

1.

Standardize Data

Used Pandas and Numpys to fairly compare employee info

2.

Build predictive models

Used Scikit-Learn and XGBoost libraries to verify 87% accuracy

3.

Detected data bias

The AI unfairly targeted groups while missing leavers

4.

Find root cause

Used SHAP to prove the AI ignored work quality and decided on unfair factors instead

2 reasons the AI model failed

3

discriminatory traits were part of the model

86%

of data were employees who stayed; the AI ignored leavers

Most over-flagged

3.7x

more young employees are predicted to quit compared to seniors

Most missed

23%

more women who quit are wrongly compared to men

Most inconsistent

50%

drop in accuracy for married employees versus single ones

Complete failures

0%

of senior leaders who quit were caught by the model

Top 3 ways to prevent future AI errors



Stop deployment

Current AI illegally uses personal details to make predictions

Risks lawsuits and reputational damage



Fix model

Rebuild the model to only use job-related factors

Reduces false alarms, saves time, and boosts morale



Build foundation for safe, fair AI

Standardize pre-deployment checks so every AI tool is tested before use

Protect the company, follow the law, and innovate responsibly

How I deliver business value in high-stakes problems

1.

I quantify risk

I audited a simulated AI tool and found it was using illegal data, like gender and age, to make decisions

If deployed, it would cost the company millions in talent and compliance

2.

I translate tech into strategy

I used tools like SHAP to show leadership why the AI model failed

This helps leaders confidently choose what resources to invest in

3.

I build systems that save time and money

I made a repeatable process to check any new technology for risks before it is used

This allows us to innovate safely without business risk

Ultimately, I solve high-stakes problems by measuring risks and building systems that protect company value.